

PORTFOLIO DECISION

SUBSTRATE ROI STRATEGY

What to ship, which formats to expose, and what the evidence does - and does not - justify.

PRIMARY COMMERCIAL EXPERIMENT	KSGP1 local ground-station pass planner: deterministic CLI, conventional orbit input, CSV events.
STRONGEST EXACT-GROVE PROOF	Editable packed-polar scene to Android APK: the retained recipe/ECS/LUT/GPU/Bayer chain.
PORTFOLIO RULE	Build a paid bounded workflow before building another general-purpose format.

EVIDENCE BOUNDARY

The scores in this report measure engineering leverage, proof maturity, distribution fit, and remaining risk. They are not forecasts of revenue or profit. The repository contains no customer, pricing, acquisition-cost, or support-cost evidence.

01 / DECISION

The portfolio answer

Three launch lanes are defensible. Only one should receive product-expansion money before a customer signal exists.

Recommendation: package the local pass planner first, preserve the exact-Grove Android path as the flagship technical proof, and use self-contained HTML5 as the lowest-friction distribution surface.

Lane	Decision	Why now	Format boundary
1 Local pass planner	Fund one bounded paid experiment	Most application-shaped verified sibling: deterministic event equality, independent orbital check, tiny seed file, clear first-query use case.	OMM/CSV -> KSGP1 -> CSV/JSON
2 Offline HTML5	Ship as a productized demo/training surface	Single-file, offline, already generated from editable project JSON; lowest adoption and installation friction.	project.json + SVG -> HTML
3 Grove Android	Use as exact-substrate flagship	Only lane exercising the retained recipe + packed ECS + LUT + instanced GPU + Bayer chain on a physical phone.	JSON -> KC packs -> APK/AAR
KSEED / GPU Native	Incubate behind named pilots	Strong integrity or kernel evidence, but product definition and/or integration remain incomplete.	KSEED + JSON; G32/G64 + E16
Universal UGTOMS	Defer	Current binaries are profile-specific and not interoperable. A universal container adds decoder, migration, and adoption cost before demand is proven.	By-reference envelope only

Critical commercial truth

- A technically high score says 'fastest route to a credible proof', not 'largest market'.
- The next missing evidence is a paid design partner, not another internal mechanism count.
- When a conventional implementation is simpler, faster, safer, or more accurate at equal behavior, use it.

02 / METHOD

What was reviewed and what was rechecked

Canonical contracts and dated machine evidence were weighted above duplicate PDFs, build logs, screenshots, and floating test counts.



Selection method

- Inventory included Markdown, text, PDF, DOCX, HTML, RST, and TeX across the Git root. Generated/cache/build-heavy paths were excluded from substantive weighting.
- The governing acceptance boundary was Grove's mechanism map: serialized meaning, reconstruction/behavior, bounded error or failure, and named evidence.
- Adjacent modules were scored as sibling profiles, not silently treated as one interoperable binary substrate.

Read-only check	Observed result	Interpretation
Grove validators	Seven 3D examples and Elizabeth 2D passed their matching validators.	Current editable formats parse and validate; this does not create new physical device evidence.
GSP4 manifest	156 files, 4,253,606 bytes, manifest valid.	Package integrity is good. The local CLI still needs optional ML dependency separation.
KSEED inspector	805,002-byte Poco log; header/chunk CRCs, hash chain, summary, and SHA-256 passed.	Integrity/replay proof is real; raw images are absent by design.
KLB 0.6 network CPU	58 objects, 16 stations, 9,335 events, zero failures; event identity passed.	Network relation pruning is CPU-verified. Its new GPU path remains open.

Score model

Evidence maturity 30% + exact-substrate fit/reuse 20% + time to credible proof 15% + distribution/interoperability 15% + differentiated avoided work 10% + inverse risk/gap 10%. Confidence describes proof quality, not demand.

03 / SUBSTRATE

The exact retained Grove slice

The repo's strongest current definition is narrow and operational. Preserving its boundaries is more valuable than broadening its name.



Exact retained Grove chain. It is a bounded profile, not a universal codec or engine.

Status	What the contract supports	What it means for product claims
Retained	Bounded SCLP log-radius chart; packed pose/motion; bounded SplitMix64 lineage; one KCPR content-addressed operator slice.	These can be named as implemented profile mechanisms.
Physical target slice	UGLUT2; Glow/Grow material field; canonical Bayer order.	One representative Grow case ran on a Poco/Mali device.
Partial / active	GPU angle/heading parity; physical derivative parity.	Do not claim complete CPU/GPU math parity.
Deferred	Metric/gradient operators; Morton packing; general content-addressed operators; multi-chart topology.	These are roadmap items, not current product capabilities.
Rejected for Grove	RGB565 Bayer-direct hot path; polar encoding for arbitrary UI, text, or topology.	Do not force the substrate into poor-fit data.

Acceptance rule: compact bytes or visual similarity are insufficient. Promotion requires serialized meaning, desktop behavior, native behavior, bounded failure/error, and a named evidence artifact.

Sibling modules are not the same wire format

KSGP1, KSEED, UGKG/UGNL, G32/G64, UG5N, and the KC packs repeat a useful architectural spine: typed state, bounded operators, deterministic gates, compact records, and evidence. They are not currently interchangeable. ROI comes from shared conformance discipline and adapters, not from pretending they are one codec.

04 / SCORECARD

Evidence-adjusted engineering ROI

The ranking favors a credible, bounded workflow with a short route to independent proof. It deliberately discounts speculative universality.



Rank	Application	Confidence	Primary reason	Primary nonclaim
1	Local ground-station pass planner	High technical / medium product	Exact event equality, independent SGP4 check, clear query-first workload.	No customer or repeated-query advantage proven.
2	Offline 2D educational/training app	High	Single-file distribution from editable data; low friction.	Does not verify the polar mobile renderer.
3	Packed-polar Android visual/microgame	High lab / medium product	Only exact-Grove physical chain proof.	Debug, one device/preset, no power or causal A/B matrix.
4	KSEED observation/evidence logger	Medium-high	Physical capture plus verified integrity chain.	Not scanner, SLAM replacement, or image compression.
5	Deterministic GPU event filter	High kernel / medium app	Strong physical sparse-event throughput and compaction.	Fixed query and separate from Grove GLES.

Scores are decision aids, not precision measurements. A five-point change should not override a clear user, integration, or safety constraint.

05 / LANE 1

Local pass planner: the first commercial experiment

This is the most application-shaped proof in the library, but its advantage exists only for the declared first-query or sparse-query workload.

Verified evidence	Observed
Compact source	KSGP1 file: 5,793 bytes for the preserved 32-object, seven-day corpus.
Independent accuracy	328 states checked; maximum preserved position delta about 0.266 mm.
Physical stress	RTX 5070 Ti Laptop; direct event path equaled dense events with no truncation in preserved runs.
First-query economics	At a 2 GiB horizon: direct p50 316.18 ms vs dense end-to-end 325.73 ms; modest 1.03x advantage.
Critical counterexample	Once dense state is resident, repeated queries can be far faster. The workload contract is part of the product.

Product shape

- User: a local operator or analyst who needs reproducible acquisition/loss windows without maintaining a dense trajectory store.
- Input: OMM/CSV plus explicit station, time basis, Earth model, and tolerance metadata.
- Internal: KSGP1 seed-chain payload and deterministic verifier.
- Output: sorted CSV for immediate use, JSON evidence receipt, and later an OpenAPI surface if the CLI contract stabilizes.

Fastest credible proof

Reproduce the preserved 717-event local-pass fixture, then run one external OMM corpus and one independent SP3/reference comparison. Measure full service latency, not kernel time only.

Kill gate: stop or reposition if a conventional orbital library is simpler, safer, and equally reproducible for the target workload, or if no paid design partner appears after 3-5 interviews.

Format stack

Layer	Format	Authority
Exchange in	OMM/OEM or documented CSV	External source authority
Compact core	KSGP1 <code>`ksgp`</code>	Profile-specific canonical payload
Exchange out	CSV + JSON manifest/receipt	Operational handoff
Distribution	Python wheel; optional OCI/OpenAPI later	Packaging, never semantic authority

06 / LANE 2

Offline HTML5: the lowest-friction product surface

Use it for bounded education, training, or interactive explanation where inspectability and zero-install distribution matter.

Best near-term application: a self-contained interactive technical explainer or educational game generated from editable project JSON and SVG assets.

WHY IT SCORES	WHAT IS PROVEN
No store, installer, native driver, or CDN is required. The artifact can be inspected, archived, and opened offline. Conventional HTML/SVG lowers buyer friction while the event/graph substrate remains reproducible behind it.	The Elizabeth vector example has canonical project JSON, a 72,161-byte generated `dist/index.html`, and a build report. The Grove CLI validators pass its 2D project. Current runtime scope includes fixed-step Canvas/vector behavior, input, audio, collision, particles, and save/load.

Proof plan

- Rebuild from canonical `project.json`; capture the output hash and build report.
- Open offline in a clean browser and exercise keyboard, gamepad, touch, audio, save, and reload.
- Run accessibility checks: keyboard-only, focus, contrast, reduced motion, text equivalents, and screen-reader landmarks.
- Test one paid use case rather than another generic demo.

Boundary

This verifies the 2D event/graph runtime and deterministic export path. It does not verify browser Mobile 3D, KCPR/UGLUT2, mobile GPU parity, or a general engine.

Authoring	Materialization	Distribution	Machine evidence
project.json + SVG	self-contained HTML/JS	single `.html`	build-report.json + hash

07 / LANE 3

Grove Android: the exact-substrate flagship

This is the only current application lane that physically traverses the retained polar recipe, packed ECS, shared LUT, instanced GPU, and Bayer chain.

Evidence field	Preserved Grow proof
Target	Poco 2412DPC0AG / Mali; Android NativeActivity + GLES3
Artifact	1,819,202-byte ARM64 debug APK; exact pack hashes recorded and byte-matched
Recipe/state	KCPR v4; 9 operators; 1 recipe; KCPK 1,690 B; KCPR 304 B; KCRP 32 B
GPU work	128 instances, 127 derived copies, one LUT, two batches, zero CPU fallback
Timing	120.33 effective FPS; p50/p95/p99 8.380/10.113/11.295 ms over the captured run
Unmeasured	No usable GPU timer, no electrical power, one device/preset, one non-causal A/B pair

Highest-ROI application shape

A small editable procedural visual, microgame, or benchmark demonstrator where hundreds of display copies share a bounded recipe and LUT. The differentiator is reproducible instanced presentation and avoided per-copy state, not generic asset compression.

Productization steps

- Start from the editable packed-polar lab scene; expose seed, recipe, material bands, instance count, and palette in authoring data.
- Create a release signing/update policy, then consider an AAR only if a customer asks for SDK integration.
- Repeat on three device/GPU classes with thermal, electrical power, and causal Direct/LUT plus Glow/Grow matrices.
- Keep KCPR members render-only; do not imply gameplay entities or physics.

Stop expansion if the customer values a conventional engine workflow more than compact procedural presentation. There is no Unity/Godot integration, GLB/skeleton pipeline, Vulkan Grove renderer, or general PCG proof.

Editable authority	Hot deployment records	Interchange	Distribution
project.json	KC3D/KCPK/KCPR/KCRP/UGLUT2	static glTF 2.0.1	APK now; AAR conditional

08 / INCUBATION

KSEED and GPU Native: strong evidence, incomplete products

Both contain real physical proof. Neither should be widened beyond its bounded semantics while its product user and integration contract remain unresolved.

	KSEED observation / evidence logger	GPU deterministic event filter
What is real	Physical Poco capture; 805,002-byte log; 8,328 observations; 348 keyframes; 8,702 events; CRC/hash chain verification; about 7.64 MiB/hour projected.	Physical RTX run at 4,194,304 candidates; G32 12.519G eval/s; about 2.2x over G64 evaluation; compact stream about 21.1x smaller at roughly 4.75% event yield.
High-ROI app	On-device benchmark, audit trail, or repeat-pass field recorder with selected external evidence.	C/C++ or service-side SDK for fixed-query sparse-event filtering with CPU oracle.
Critical defect	23.29% camera arrivals overwritten in the preserved run; no retained raw images; no trusted time/signature.	Fixed query, capability-specific records, FP16 predicate changes, readback/transfer economics, separate from Grove GLES.
Do not claim	Scanner, image codec, VIO/SLAM replacement, depth, metric reconstruction, loop closure, or forensic authenticity.	Universal speedup, general GPU database, or transfer of Vulkan/CUDA results to Grove.
Next proof	Fix scheduling; add annotations, trusted time/location, optional encrypted selected images, hardware signing/external anchor; run one field workflow.	Publish a minimal ABI, fixtures, fail-closed overflow, full end-to-end benchmarks, and one real sparse-event workload.

Hashes prove byte integrity and linkage. They do not, by themselves, prove who captured data, when it was captured, authorization, or ownership.

Format recommendation

- Keep `.kseed`` and G32/G64/E16 profile-specific and independently verifiable.
- Expose JSON/JSONL receipts and conventional materializations. Use GeoPackage for offline field handoff and Parquet/GeoParquet for analytics.
- Use PLY only as a downstream geometry derivative when reconstruction actually occurred; it is never the canonical KSEED log.

09 / OPTIONS

High-upside modules that are not product-ready

These are legitimate experiments only when the listed missing evidence is funded explicitly. They should not inherit claims from better-proven siblings.

Option	Why it could matter	Missing before promotion	Decision
Multi-station scheduler	Direct extension of the strongest application-shaped lane.	Resource conflicts, link budgets, fresh data, Earth-orientation scope, UI, and v0.6 physical GPU proof.	Next after one pass-planner partner.
By-reference UGTOMS envelope	Highest repo-wide conformance leverage without changing payload bytes.	Common profile schema, two independent readers, four exact round trips, malformed/unknown fail-closed cases.	Build after P0; no universal transcode.
GSP4 spatial gate	Plausible sparse asset/proximity/event workflow.	Physical GPU closure, slim core packaging, real domain data, and proposer value above current ~0.55 smoke accuracy.	Pilot only.
Knowledge conduit	Typed claims, deterministic gates, UNKNOWN path, optional non-authoritative LLM.	One bounded manual, ingestion/adapters, provenance schema, evaluation with technicians/auditors.	Concept pilot.
Definition Breaker	Strong accessibility/learning design and offline format fit.	Executable prototype plus participatory child, disability, education, and assistive-tech validation.	Social-impact experiment.
3MF/STEP, GLB/KTX2 adapters	Useful cross-domain proof against established standards.	Implement, round-trip, quantify residuals/errors, and identify a buyer workflow.	P3 only.

Explicit low-ROI or high-risk directions now

- General-purpose `.ugtoms`` codec/container; general PCG or game engine; full SLAM/scanner; safety-critical orbital service.
- SARA secret recovery, signing, or wallet product. The implemented package is deliberately public/watch-only conformance.
- Go or Chess solution claims. Full roots remain UNKNOWN or UNRESOLVED.
- 4D, full OpenUSD, GLB/skeletal pipeline, Unity/Godot, and Vulkan Grove integration without a customer reason.

10 / FORMATS

Use standards at the edge; keep custom binaries in the core

The highest-ROI file strategy minimizes adoption cost while preserving the packed records that actually deliver deterministic execution or avoided materialization.

Need	Use now	Role	Decision / caveat
Contracts/config	JSON + JSON Schema 2020-12	Editable and exchange authority	Version, validate, and declare canonicalization per profile.
Events/receipts	CSV + JSONL	Simple operations + append/replay	Always state order, units, time basis, tolerance, and failure state.
Provenance	JSON-LD 1.1 + W3C PROV mapping	Portable linked evidence	Use at exchange boundary; do not force RDF into hot records.
Spatial offline	GeoPackage 1.4	Portable SQLite mobile handoff	Good fit for KSEED/field workflows.
Spatial analytics	GeoParquet 1.1 + Parquet	Columnar analysis	Pin stable 1.1 rather than a release candidate.
2D/browser	SVG + self-contained HTML	Inspectability and zero-install distribution	Keep canonical project JSON and build evidence beside it.
3D	glTF 2.0.1 JSON now	Static presentation interchange	No GLB import or animation pipeline is implemented.
Orbit	OMM/OEM/CSV; SP3 reference	Conventional domain boundary	KSGP1 stays internal; identify model/frame/time profile.
Distribution	Wheel, APK/AAR, optional OCI/OpenAPI	Installation/deployment	Containers are not semantic authority; releases need signatures.
Human record	PDF	Review and audit view	Never the sole machine authority.

Internal formats to preserve, not universalize

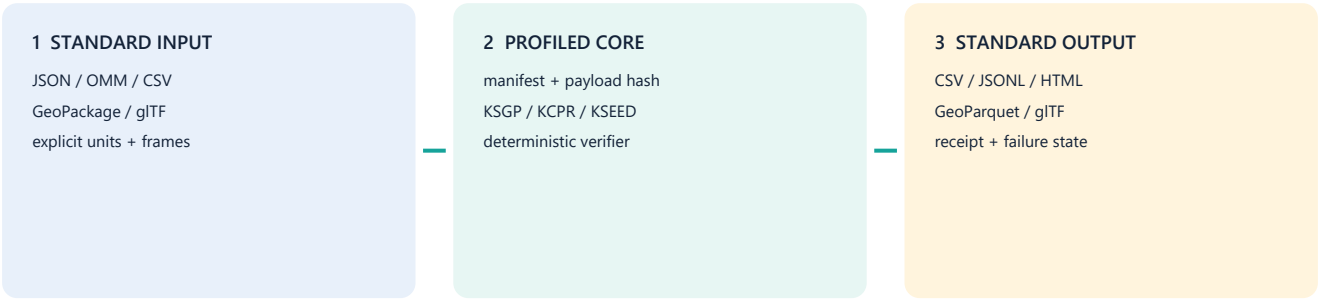
Grove: KC3D392, UGECS1, UGLUT2, KCPK392, KCPR392, KCRP392, KCVG001, KCAN392, KCHI392, KCSP392. Siblings: KSGP1, KSEED, UGKG/UGNL/UGDEPLOY, G64/G32/G24, E16/E32, UG5N. Each needs a profile owner, versioned schema, fixtures, verifier, independent reader, limits, and migration/failure policy.

Do not compute a whole-product compression ratio by dividing recipe bytes by baked output bytes while omitting mesh, texture, decoder, residual, or reconstruction cost.

11 / ARCHITECTURE

A small conformance envelope, not a universal transcode

The envelope should identify and verify profile-specific payloads by reference. It must not pretend semantic equivalence across domains.



The standards boundary carries adoption. The custom payload carries compact execution and proof.

Required field	Purpose
profile_id + component_version_id	Select exact semantics; prevent bare-version and mechanism collisions.
payload URI/bytes + SHA-256	Preserve original payload identity without transcoding.
units, frames, time basis, quantization	Make reconstruction interpretable and error claims auditable.
capability requirements + limits	Fail closed on unsupported GPU/features/caps.
acceptance contract + verifier command	Name the behavior/error gate and reproducible check.
source lineage + residual/literal reference	Keep provenance and irreducible data outside the recipe fantasy.
failure state	UNKNOWN/rejected/overflow/unsupported must be first-class outputs.

Promotion gates

- Wrap KCPR, KSGP1, KSEED, and one GPU record corpus without changing their bytes.
- Build two independent readers and exact round-trip fixtures.
- Reject malformed and unknown profiles deterministically.
- Materialize a conventional output for each profile and compare at its declared tolerance.
- Adopt a universal extension only after two unrelated domains and two independent implementations use the envelope.

12 / ROADMAP

Ninety days to a decision, not ninety days of scope

Resolve commercial blockers first, then prove one bounded workflow against a conventional baseline and a real buyer signal.

Window	Outcome	Deliverables	Gate
Days 0-14	The repo can be discussed externally without ambiguous rights, identity, or floating evidence.	Rights inventory; owner/counsel license decision; redacted report; component registry; evidence registry; stale-status reconciliation; format registry.	Named owner and resolution date for every red item.
Days 15-30	Local pass planner installs and reproduces its preserved fixture.	Slim CLI/wheel; OMM/CSV input; KSGP1 verification; 717-event CSV; manifest; baseline harness; explicit workload contract.	Clean install without ML/GPU extras; exact sorted events.
Days 31-45	External technical evidence and buyer evidence exist.	External OMM corpus; independent SP3/reference check; full-service p50/p95/memory; 3-5 interviews; paid-pilot proposal.	Equal behavior/error and one credible budget owner.
Days 46-60	Exact-Grove proof is repeatable and editable.	Scene-driven Grow demo; release-signing plan; multi-device/power matrix protocol; causal A/B plan.	No hidden bootstrap/code-only scene; zero fallback.
Days 61-75	A real workflow is funded or the lane stops.	Paid design partner, operational acceptance criteria, support estimate, deployment choice.	Money or explicit internal budget; otherwise reposition.
Days 76-90	One portfolio lane is selected.	Decision record: scale pass planning, pivot to evidence logging, or remain a conformance/benchmark library.	Other lanes explicitly deferred.

The day-75 gate is deliberately commercial. More internal tests cannot substitute for a buyer, operational acceptance criteria, and cost-to-deliver evidence.

Decision metrics

- Correctness: exact event identity or declared bounded error against an independent reference.
- Economics: end-to-end latency, memory, storage, energy, install/support effort, and conventional baseline.
- Adoption: clean install, standards input/output, independent reader, and one named user workflow.
- Commercial: design-partner budget, price hypothesis, support burden, and stop criterion.

13 / BLOCKERS

What can invalidate the strategy

Several current blockers are more material than any missing opcode or file extension.

Risk	Status	Why it matters	Required action
Root licensing / rights	RED	Submodule licenses do not create a corpus-wide commercial grant; no top-level LICENSE/NOTICE exists.	Rights inventory and owner/counsel decision before public/commercial distribution.
Private identity in draft	RED	The cross-domain draft contains personal data unsuitable for public release.	Redact and scan every release candidate.
Version/mechanism collisions	RED	Bare 3.9.x/4.x and M-number references are not globally unique.	Use component-scoped IDs and refresh the registry.
Evidence drift	AMBER	Older Grove, KSEED, and KLB status text contradicts later dated artifacts.	Evidence registry with chronology and supersession.
Dependency coupling	AMBER	GSP4 inspection currently pulls optional PyTorch in the local environment.	Split deterministic core/inspectors from ML extras.
Physical scope	AMBER	Grove has one phone/preset; KLB 0.6 GPU open; GSP4 GPU open; KSEED overwrite high.	Close only the lane-specific device gates.
Format proliferation	AMBER	Each bespoke binary creates reader, migration, tooling, and adoption obligations.	Freeze registry; no new format without a paid workflow and independent reader plan.

Hard nonclaims

No universal codec; no general engine; no complete SLAM/scanner; no raw-image compression in KSEED; no KSGP advantage for every query pattern; no hash-only authenticity; no solved Go/Chess; no high-stakes wallet; no 4D implementation; no full GLB/skeleton/OpenUSD pipeline; no Unity/Godot integration.

14 / EVIDENCE

Local evidence register

Paths are relative to the parent Git root. Dated physical artifacts outrank undated summaries; profile boundaries stay attached to their originating component.

ID	Claim supported	Primary local source
E-GROVE-DEF	Exact Grove chain, retained/deferred/rejected statuses, acceptance rule	UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES/docs/UGTS_SUBSTRATE_MECHANISM_MAP.md
E-GROVE-ARCH	Editable/native flow, KC formats, limits, browser/mobile scope	UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES/docs/ENGINE_ARCHITECTURE.md
E-GROVE-PHYS	Poco Grow install/launch/profile, pack hashes, timings, limits	UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES/build/release-handoff/20260830T023441Z-polar-grow/evidence.json
E-KSGP-PHYS	Physical direct-vs-dense event equality and independent SGP4 comparison	klb_seedchain_gpu_v0.5.0_full_sgp4/.../verification_sgp4_2gib_console.txt and verification_independent_sgp4_summary.json
E-KLB-NET	v0.6 network relation pruning CPU proof; new GPU acceptance open	klb_seedchain_gpu_v0.6.0_network_pruning/.../README.md and DELIVERY_STATUS.md
E-KSEED	Physical observation log metrics and integrity boundary	UGTS_KC_4_1_Poco_X7_Pro_Seed_Native_Package/.../validation/device_poco_x7_pro_2026-08-28/
E-GPU	Physical RTX G64/G32 evaluation and sparse-event compaction	Unified_Geometric_Topological_Substrate_GPU_Native_Package/.../benchmarks/PHYSICAL_GPU_REPORT_RTX_5070_T1_LAPTOP.md
E-GSP4	CPU manifest/graph proof, G32 error, ML smoke, GPU open	gsp4_spatial_kd_v0.5.0/gsp4_spatial_kd_v0.5.0/docs/VALIDATION.md
E-CROSS	Cross-domain boundary, standards crosswalk, rights gaps	UGTOMS_cross_domain_pilot_overview_review_draft.pdf
E-VERSION	Component IDs, version and mechanism collisions	UGTS_Versioning_Charter_Phase_2_Library_Wide_Registry.pdf

Evidence cautions

- Grove's full Direct/LUT and Glow/Grow matrices remain incomplete; one physical Grow case cannot establish broad causal performance.
- KLB 0.5 physical GPU evidence does not close KLB 0.6 network-GPU acceptance.
- KSEED 4.1 physical evidence does not transfer to Atlas 4.1.1, which documents a clean reimplementaion from the report.
- GSP4's package manifest and CPU validation do not prove operational geospatial value or GPU closure.

15 / REFERENCES

External standards used for the format strategy

These standards are boundary recommendations. Their presence in this report is not a claim that every adapter is already implemented.

Standard	Why it is recommended	Official source
JSON Schema 2020-12	Versioned machine contracts and validation	json-schema.org/draft/2020-12
JSON-LD 1.1	Portable linked semantics without changing hot records	w3.org/TR/json-ld11
W3C PROV	Interoperable provenance concepts and mappings	w3.org/TR/prov-overview
OGC GeoPackage 1.4	Portable SQLite spatial handoff for mobile/offline use	OGC 12-128r19
GeoParquet 1.1	Stable columnar geospatial analytics interchange	geoparquet.org/releases/v1.1.0
Apache Parquet	Efficient columnar analytical storage	parquet.apache.org/docs/file-format
glTF 2.0.1	Established static 3D presentation interchange	registry.khronos.org/glTF
CCSDS Orbit Data Messages	Conventional orbit exchange boundary	CCSDS 502.0-B-3
Python wheel	Reproducible CLI/library distribution	Python Packaging specification
Android AAR	Reusable Android library distribution when demanded	Android library guide
OpenAPI	Service contract after the CLI/API semantics stabilize	OpenAPI specification

Regulatory pull, not a legal conclusion

Traceable logs, documentation, human oversight, and product provenance are increasingly useful design properties. Applicability depends on the actual product and role. This report is not legal advice; use the final product scope to determine obligations.

EU AI Act (Regulation 2024/1689) | EU Ecodesign for Sustainable Products Regulation 2024/1781

Final decision: standards at the edges, profile-specific deterministic records in the core, and one paid bounded workflow before any new universal format.